

Derivation of the $q(L)$ Variational Update

Supervised Bayesian Matrix Factorization

Self-Contained Reference for `code/update_L.R`

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Abstract

This document derives the coordinate-ascent variational inference (CAVI) update for the patient loading matrix L in the Supervised Bayesian Matrix Factorization model. Unlike the β update (a one-dimensional EBNM problem), the L update is a *vector* EBNM because the precision $A_L[i]$ and signal $B_L[i]$ vary across patients i — the Cox diagonal Hessian weights W_{ii} are patient-specific. The derivation shows that l_{ik} receives signal from two independent sources: (1) the genomics residual (weighted by τ_j) and (2) the survival working response (weighted by W_{ii}). The result is an n -observation EBNM problem solved once per factor k per CAVI sweep, using the `ebnm` R package with a `point_normal` prior.

This is the second of four modular derivation documents ($q\beta \rightarrow qL$ (this) $\rightarrow qF \rightarrow q\tau$).

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1 Setup and Notation

1.1 Model Summary

$$\text{Genomics: } Y_{ij} = \sum_{k=1}^K l_{ik} f_{jk} + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \tau_j^{-1}) \quad (1)$$

$$\text{Survival: } \eta_i = \sum_{k=1}^K l_{ik} \beta_k = \mathbf{l}_i^\top \boldsymbol{\beta}, \quad h(t_i) = h_0(t_i) \exp(\eta_i) \quad (2)$$

1.2 Key Notation

Symbol	Definition	R variable
$\bar{l}_{ik}$	$\mathbb{E}_q[l_{ik}]$	<code>EL[i,k]</code>
$\bar{l}_{ik}^2$	$\mathbb{E}_q[l_{ik}^2] = \text{Var}_q + \bar{l}_{ik}^2$	<code>EL2[i,k]</code>
$\bar{f}_{jk}$	$\mathbb{E}_q[f_{jk}]$	<code>EF[j,k]</code>
$\bar{f}_{jk}^2$	$\mathbb{E}_q[f_{jk}^2]$	<code>EF2[j,k]</code>
$\bar{\beta}_k$	$\mathbb{E}_q[\beta_k]$	<code>EBeta[k]</code>
$\bar{\beta}_k^2$	$\mathbb{E}_q[\beta_k^2]$	<code>EBeta2[k]</code>
W_{ii}	Cox Hessian diagonal (positive)	<code>w[i]</code>
z_i	Cox working response	<code>z[i]</code>
z_i^{-k}	Partial working response (excl. factor k)	<code>z_no_k[i]</code>
R_{ij}^{-k}	Partial residual matrix	<code>R_k[i,j]</code>

1.3 Partial Residual Matrix R^{-k}

When updating factor k , the residual excluding the contribution of that factor is:

$$R_{ij}^{-k} = Y_{ij} - \sum_{k' \neq k} \bar{l}_{ik'} \bar{f}_{jk'} = Y_{ij} - \mathbf{l}_i^\top \mathbf{f}_j + \bar{l}_{ik} \bar{f}_{jk} \quad (3)$$

In matrix form: $R^{-k} = Y - EL EF^\top + \text{outer}(EL[,k], EF[,k])$.

1.4 Partial Working Response z^{-k}

Similarly, the Cox working response excluding factor k :

$$z_i^{-k} = z_i - \bar{\beta}_k \bar{l}_{ik} \quad (4)$$

This is the contribution to z_i from all factors *except* k .

2 CAVI Objective for $q(l_{ik})$

The ELBO with respect to $q(l_{ik})$ (holding all other variational posteriors fixed) contains two relevant terms:

$$\mathcal{F}[q(l_{ik})] \supset \underbrace{\mathbb{E}_q[\log P(Y | L, F, \tau)]}_{\text{genomics}} + \underbrace{\mathbb{E}_q[\log P(t, \delta | L, \boldsymbol{\beta})]}_{\text{survival}} + \mathbb{E}_{q_{l_k}} \left[\log \frac{g_{l_k}}{q_{l_k}} \right] \quad (5)$$

where the KL term $\mathbb{E}[\log g/q]$ is handled implicitly by the EBNM solver. We focus on expanding both likelihood terms with respect to l_{ik} .

3 Genomics Contribution to A_L and B_L

3.1 Extracting the Quadratic in l_{ik}

The genomics log-likelihood for feature j and patient i is:

$$\log P(Y_{ij} \mid l_{ik}, \dots) \propto -\frac{\tau_j}{2} \mathbb{E}_q \left[(Y_{ij} - R_{ij}^{-k} - l_{ik} f_{jk})^2 \right] \quad (6)$$

where we have substituted $Y_{ij} - R_{ij}^{-k} = \bar{l}_{ik} \bar{f}_{jk}$ (the within-factor contribution at posterior means) and now treat l_{ik} as the variable. Expanding:

Derivation Steps

$$\begin{aligned} \mathbb{E}_q \left[(R_{ij}^{-k} - l_{ik} f_{jk})^2 \right] &= (R_{ij}^{-k})^2 - 2 l_{ik} R_{ij}^{-k} \mathbb{E}_q[f_{jk}] + l_{ik}^2 \mathbb{E}_q[f_{jk}^2] \\ &= (R_{ij}^{-k})^2 - 2 l_{ik} R_{ij}^{-k} \bar{f}_{jk} + l_{ik}^2 \bar{f}_{jk}^2 \end{aligned}$$

Summing over all features $j = 1, \dots, p$ and multiplying by $-\tau_j/2$:

$$\text{genomics contribution} = -\underbrace{\frac{l_{ik}^2}{2} \sum_j \tau_j \bar{f}_{jk}^2}_{A_{L,\text{gen}}} + l_{ik} \underbrace{\sum_j \tau_j R_{ij}^{-k} \bar{f}_{jk}}_{B_{L,\text{gen}}[i]} + \text{const} \quad (7)$$

Key observations:

- $A_{L,\text{gen}} = \sum_j \tau_j \bar{f}_{jk}^2$ is a *scalar* — the same for all patients i . It uses the full second moment $\bar{f}_{jk}^2$ (not $\bar{f}_{jk}^2$) — this is the error-in-variables correction.
- $B_{L,\text{gen}}[i] = \sum_j \tau_j R_{ij}^{-k} \bar{f}_{jk}$ is an n -vector, equal to the i -th entry of $(R_k)(\tau \odot EF_k)$ where $\tau \odot EF_k$ denotes element-wise multiplication.

Code Mapping (update.L.R)

```
A_gen <- sum(Tau * EF2_k)           # scalar
B_gen <- as.vector(R_k %*% (Tau * EF_k)) # n-vector
```

4 Survival Contribution to A_L and B_L

4.1 Taylor Approximation of the Cox Likelihood

The Cox partial-likelihood is non-Gaussian. The standard treatment (as in the parent document) applies a second-order Taylor expansion around the current linear predictor $\hat{\eta}_i$. The resulting approximation has the form of a weighted least-squares objective:

$$\log P(t_i, \delta_i \mid \mathbf{l}_i, \boldsymbol{\beta}) \approx -\frac{W_{ii}}{2} (\eta_i - z_i)^2 + \text{const} \quad (8)$$

where $W_{ii} > 0$ is the diagonal element of the negative Hessian and z_i is the working response.

4.2 Quadratic in l_{ik}

Substituting $\eta_i = l_{ik}\beta_k + z_i^{-k}$ (where z_i^{-k} collects all other factors):

Derivation Steps

$$-\frac{W_{ii}}{2}\mathbb{E}_q\left[(l_{ik}\beta_k + z_i^{-k} - z_i)^2\right] = -\frac{W_{ii}}{2}\mathbb{E}_q\left[(l_{ik}\beta_k - \underbrace{(z_i - z_i^{-k})}_{\approx \bar{\beta}_k \bar{l}_{ik}})^2\right]$$

Expanding and collecting terms in l_{ik} :

$$\text{survival contribution} = -\frac{l_{ik}^2}{2} \underbrace{W_{ii} \bar{\beta}_k^2}_{A_{L,\text{surv}}[i]} + l_{ik} \underbrace{W_{ii} z_i^{-k} \bar{\beta}_k}_{B_{L,\text{surv}}[i]} + \text{const} \quad (9)$$

Key observations:

- $A_{L,\text{surv}}[i] = W_{ii} \bar{\beta}_k^2$ is an n -vector (varies across patients through W_{ii}). It uses the full second moment $\bar{\beta}_k^2$, not $\bar{\beta}_k^2$.
- $B_{L,\text{surv}}[i] = W_{ii} z_i^{-k} \bar{\beta}_k$ is also an n -vector.

Code Mapping (update.L.R)

```
A_surv <- w * EBeta2_k          # n-vector
B_surv <- w * z_no_k * EBeta_k  # n-vector
```

5 Combined Update: Vector EBNM Problem

5.1 Combining Both Sources

Adding the genomics and survival contributions (equations 7 and 9), the objective for $q(l_{ik})$ is:

$$\mathcal{F}[q(l_{ik})] \supset -\frac{l_{ik}^2}{2} \underbrace{(A_{L,\text{gen}} + A_{L,\text{surv}}[i])}_{A_L[i]} + l_{ik} \underbrace{(B_{L,\text{gen}}[i] + B_{L,\text{surv}}[i])}_{B_L[i]} + \mathbb{E}_{q_{l_k}} \left[\log \frac{g_{l_k}}{q_{l_k}} \right] \quad (10)$$

where:

$$A_L[i] = \sum_j \tau_j \bar{f}_{jk}^2 + W_{ii} \bar{\beta}_k^2 \quad (11)$$

$$B_L[i] = \sum_j \tau_j R_{ij}^{-k} \bar{f}_{jk} + W_{ii} z_i^{-k} \bar{\beta}_k \quad (12)$$

5.2 EBNM Formulation

Completing the square in l_{ik} and recognising the Gaussian pseudo-likelihood:

$$q^*(l_{ik}) \propto g_{l_k}(l_{ik}) \mathcal{N}\left(l_{ik} \mid \underbrace{B_L[i]/A_L[i]}_{x_i}, \underbrace{1/A_L[i]}_{s_i^2}\right) \quad (13)$$

This is an n -dimensional EBNM problem: for each factor k , all n patients share the same prior g_{l_k} but each patient i contributes one pseudo-observation $x_i = B_L[i]/A_L[i]$ with pseudo-noise $s_i = 1/\sqrt{A_L[i]}$.

EBNM Problem

EBNM problem for $q(l_k)$:

$$\text{EBNM}(x_i = B_L[i]/A_L[i], s_i = 1/\sqrt{A_L[i]}, g = g_{l_k}) \quad i = 1, \dots, n \quad (14)$$

Calling convention: `ebnm(x = B_L/A_L, s = 1/sqrt(A_L), prior_family = "point_normal")`

Returns: `posterior$mean` $\rightarrow \bar{l}_{ik}$, `posterior$sd` $\rightarrow \sigma_{ik}$, $\bar{l}_{ik}^2 = \sigma_{ik}^2 + \bar{l}_{ik}^2$

Code Mapping (update.L.R)

```
A_L <- pmax(A_gen + A_surv, A_floor)      # n-vector [floored]
B_L <- B_gen + B_surv                     # n-vector
res <- ebnm(x = B_L / A_L, s = 1/sqrt(A_L), prior_family = prior_family)
l_mean <- res$posterior$mean              # n-vector: E_q[l_{ik}]
l_sd <- res$posterior$sd                  # n-vector: SD_q[l_{ik}]
l_second <- l_sd^2 + l_mean^2              # n-vector: E_q[l_{ik}]^2
```

6 Error-in-Variables Interpretation

The use of $\overline{f_{jk}^2} = \text{Var}_q(f_{jk}) + \bar{f}_{jk}^2$ (rather than $\bar{f}_{jk}^2$) in $A_{L,\text{gen}}$ is the *error-in-variables* correction for uncertain factors:

- If we used $\bar{f}_{jk}^2$, we would treat the factor column as known. Any posterior uncertainty in f_{jk} would be ignored.
- Using $\overline{f_{jk}^2}$ properly inflates the effective precision $A_{L,\text{gen}}$, causing the posterior of l_{ik} to shrink more. This prevents loadings from chasing poorly estimated factors.

Similarly, $\overline{\beta_k^2}$ (not $\bar{\beta}_k^2$) inflates $A_{L,\text{surv}}$.

Verification Check

Monotonicity check. For fixed signal B_L , larger A_L means: (a) smaller pseudo-noise $s_i = 1/\sqrt{A_L[i]}$, (b) same pseudo-obs $x_i = B_L[i]/A_L[i]$ but interpreted as more precise, (c) EBNM shrinks less (posterior mean $\rightarrow x_i$) but posterior SD is smaller. Net effect: tighter posterior — not necessarily more shrinkage.

7 Gauss-Seidel Ordering in update_L_all

The wrapper `update_L_all` loops over $k = 1, \dots, K$ in order. After updating column k , the new $\bar{l}_{ik}$ and $\bar{l}_{ik}^2$ are immediately used when computing R^{-k} for $k+1, \dots, K$. This Gauss-Seidel (sequential) CAVI approach converges faster than the Jacobi (synchronous) alternative.

Code Mapping (update_L.R)

```
for (k in seq_len(K)) {
  R_k      <- compute_R_k(Y, EL_curr, EF, k)    # uses updated EL[,1..k-1]
  res_k    <- update_L_k(Tau, EF[,k], EF2[,k], w, EBeta[k], EBeta2[k],
                        R_k, z_no_k)
  EL_curr[, k] <- res_k$mean
  EL2_curr[, k] <- res_k$second
}
```

8 Verification Checks

Verification Check

V1: Genomics-only limit. When $W_{ii} = 0$ for all i , the survival term drops out. $A_L[i] = A_{L,\text{gen}}$ (constant across i), and $B_L[i] = B_{L,\text{gen}}[i]$. All patients share the same posterior precision — the update reduces to a weighted regression on the factor column.

V2: Survival-only limit. When $\tau_j = 0$ for all j (or $p = 0$), $A_{L,\text{gen}} = 0$, so $A_L[i] = W_{ii}\bar{\beta}_k^2$. The update uses only the survival signal.

V3: Second moment non-negativity. $\bar{l}_{ik}^2 = \sigma_{ik}^2 + \bar{l}_{ik}^2 \geq \bar{l}_{ik}^2$, with equality iff the posterior is degenerate. In code: `min(EL2 - EL^2) >= 0` (up to floating-point tolerance).

V4: Floor prevents degeneracy. `pmax(A_L, A_floor)` ensures $A_L > 0$ even when $\tau_j = 0$ and $W_{ii} = 0$. Without the floor, $s_i = 1/\sqrt{0} = \infty$ and the EBNM call would fail.

9 Code Mapping

Code Mapping (update_L.R)

Function `update_L_k` in code/update_L.R:

```
update_L_k <- function(Tau, EF_k, EF2_k, w, EBeta_k, EBeta2_k,
                      R_k, z_no_k,
                      prior_family = "point_normal", A_floor = 1e-10) {
  A_gen <- sum(Tau * EF2_k)                # scalar
  A_surv <- w * EBeta2_k                  # n-vector
  A_L <- pmax(A_gen + A_surv, A_floor)     # n-vector [floored]
  B_gen <- as.vector(R_k %*% (Tau * EF_k)) # n-vector
  B_surv <- w * z_no_k * EBeta_k          # n-vector
  B_L <- B_gen + B_surv                   # n-vector
  res <- ebnm(x = B_L / A_L, s = 1/sqrt(A_L), prior_family = ...)
  list(mean = res$posterior$mean,
```

```

    second = res$posterior$sd^2 + res$posterior$mean^2, ...)
}

```

— Equation — Code line — Variable — $A_{L,\text{gen}} = \sum_j \tau_j \overline{f_{jk}^2}$
— `sum(Tau * EF2_k)` — scalar — $A_{L,\text{surv}}[i] = W_{ii} \overline{\beta_k^2}$ — `w * EBeta2_k` — n -vector — $B_{L,\text{gen}}[i]$
— `R_k %*% (Tau * EF_k)` — n -vector — $B_{L,\text{surv}}[i]$ — `w * z_no_k * EBeta_k` — n -vector —
 x_i — `B_L / A_L` — n -vector — s_i — `1/sqrt(A_L)` — n -vector —

10 Algorithm Summary

1. For $k = 1, \dots, K$ (Gauss-Seidel order):
 - (a) Compute partial residual: $R^{-k} = Y - EL EF^\top + \text{outer}(EL[, k], EF[, k])$
 - (b) Compute partial working response: $z_i^{-k} = z_i - \bar{\beta}_k \bar{l}_{ik}$
 - (c) Genomics precision (scalar): $A_{L,\text{gen}} = \sum_j \tau_j \overline{f_{jk}^2}$
 - (d) Survival precision (n -vector): $A_{L,\text{surv}}[i] = W_{ii} \overline{\beta_k^2}$
 - (e) Combined precision: $A_L[i] = \max(A_{L,\text{gen}} + A_{L,\text{surv}}[i], \epsilon)$
 - (f) Signal (n -vector): $B_L[i] = \sum_j \tau_j R_{ij}^{-k} \bar{f}_{jk} + W_{ii} z_i^{-k} \bar{\beta}_k$
 - (g) Solve EBNM: $\text{EBNM}(x_i = B_L[i]/A_L[i], s_i = 1/\sqrt{A_L[i]})$
 - (h) Update $\bar{l}_{ik} \leftarrow$ posterior mean, $\bar{l}_{ik}^2 \leftarrow \sigma_{ik}^2 + \bar{l}_{ik}^2$

A Notation Index

Symbol	Meaning	Dimensions
Y	Genomics data	$n \times p$
L	Patient loadings	$n \times K$
F	Biological factors	$p \times K$
β	Survival coefficients	K -vector
τ_j	Feature noise precision	p -vector
W_{ii}	Cox Hessian diagonal	n -vector
z_i	Cox working response	n -vector
R^{-k}	Partial residual (excl. k)	$n \times p$
$A_L[i]$	EBNM precision for patient i	n -vector
$B_L[i]$	EBNM signal for patient i	n -vector